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ORIGINAL RESEARCH

Lending a Hand in Bringing Ballcarriers Down: An Extended Study of Individual Defensive Contributions in the National Football League (NFL)

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Abstract
In this paper we examine game data from the National Football League (NFL) to uncover insights on how defensive teammates collectively bring down offensive ballcarriers. Traditional defensive statistics focus on individual tacklers, which is certainly vital, but in a team game more than one defender often contributes to a stop. We found that individuals who slowed down ballcarriers had a strong positive correlation with tackles. We also found that the distance that defensive players were from ballcarriers was a statistically significant predictor of ballcarrier speed. We were particularly interested in the influence that defender distances had on ballcarrier speed deceleration, leading to a tackle. We conclude with recommendations for coaches, agents, and front office executives. This paper aims to contribute to the sports analytics topics of player evaluation and performance analysis using player-tracking data, regression modeling, and data science techniques.

Keywords Individual contributions · NFL defender · Gridiron football · American football · Sports analytics · Player-tracking · Data science · Multivariate regression

Introduction

As National Football League (NFL) fans watch from the stands they are often begging or screaming for defenders on their team to read and recognize the play right away. The quicker the better, so that offensive ballcarriers can be surrounded before picking up a head of steam headed down the field to score six points. Those fans loudly celebrate the individuals that make the final tackle.

Defense strategies are designed by coaches to get those tackles and thereby prevent ballcarriers from gaining yards. If an offensive player has gained possession of the ball and is making positive yardage gains, they need to be slowed down as soon as possible. By analyzing spatiotemporal data, we can quantify the importance of defender proximity to ballcarriers. We give specific insights into how much defensive players impact the speed of the ballcarrier and which players do it best. For those studying game film, high-impact defenders will jump off the screen with flashes of brilliance between the sidelines, but what about unheralded players who do the little things right?

These players deserve credit too. Their contributions might not be spotted on a consistent basis since their actions may not result in a tackle for themselves, but it may help set one up for a teammate. Teams currently tally tackles and assisted tackle metrics, but as noted by Nguyen, Jiang, Ellingwood, and Yurko: "Surprisingly, tackles are not an official statistic in the National Football League… During each game, the home team score keeper is tasked with tracking whether a player records a tackle." [1]. Gang tackles, which occur when more than one person contributes, can make tallying even more challenging. Due to the amount of fast-moving bodies involved, making a judgement call on

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who made the tackle and who assisted is a tough ask for the home statisticians.

This paper aims to go deeper than the act of tackling. Instead, our quantitative approach is interested in the ballcarrier's entire journey that led to the point where they could be brought down rather than just the final action of the tackle. Therefore, our goal is to add a new variable that coaches can account for in their statistical evaluations aside from just tackles and assisted tackles.

After all, the objective of every team's defense as a whole is to work together to bring ballcarriers down as soon as possible, saving every yard[1] they can, and keeping ballcarriers out of the end zone[2] on each play. Many players are taught since their youth playing days that even if they cannot make an individual tackle themselves, as long as they slow a ballcarrier down until their teammates arrive that they have fulfilled their duties and contributed.

We show that by analyzing ballcarrier speed and defender distance data we can assign credit to individual defenders who influence the play. By doing so we can identify who can be credited with the largest deceleration on any given play. These individual credits for slowing ballcarriers down will be referred to as *slowdowns*. The foundational data was provided as part of the 2024 NFL Big Data Bowl competition on Kaggle [2]. This competition provided NFL Next Gen Stats player tracking data and Pro Football Focus (PFF) data from the 2022 NFL season, weeks 1–9. This paper is adapted from our submission [3] for the coaching presentation track, with the purpose of augmenting coaches' defensive player evaluation as an added layer to their film study throughout the season.

In the next section we provide our approach to address the current gap in gridiron football defensive statistics. We outline our research framework to "tackle" the current problem by introducing our *slowdown* metric to accurately capture and quantify defender influence on ballcarrier speed.

[1] Gridiron [4], or American [5], football is typically played in the United States of America, which is not on the metric system. Instead, yards (0.914 m) are used to measure the field of play. The distance between the end zones is 100 yards. Millions of people in the world use "football" for the game with the ball that is kicked and goes by "soccer" in the USA. To avoid confusion, we will use "gridiron football" for this paper.

[2] The end zone [6], is where teams aim to end up on each offensive possession. This is where touchdowns are scored, resulting in six points for the offense which is the most points possible on any singular play.

Research Framework

The challenge in focusing in on defenders who did not record a tackle or an assisted tackle on a particular play is that without tracking data, it would be as if those players never existed at all—at least on the stat sheet. Solely reviewing box score[3] data might give a misleading view of which defenders were the most vital in a game. Since modern defensive strategy involves some players having roles where they "funnel" or lead offensive ballcarriers to the rest of the defensive reinforcements, capturing their influence on the game was of key interest to us.

Even if players do not have a defined role to funnel, to "contain", or to "set the edge", they still might impact the path and speed of the ballcarrier by redirecting them in some way, opening the door for others to make the tackle. It is simply a matter of getting in the ballcarrier's way or being a possible threat to make a tackle. These scenarios make a ballcarrier think twice about their plan and that influence is what we aim to quantify.

We inherently recognize the importance of defenders swarming ballcarriers, driven by the playing experience of one of our authors. However, we also recognize the importance of quantifying the relationship between the two entities so that the conventional coaching wisdom has statistical ground to stand on. We were confident that our quantitative investigation would reinforce this belief, as defenses are designed to cut off angles and space for ballcarriers to navigate freely. Therefore, our research questions were as follows:

Research Questions

1. Can individual tracking data be used to reliably assign *slowdown* credits?
2. Do *slowdowns* correlate with traditional measures of defensive success?
3. Does defender proximity predict ballcarrier speed?

Hypotheses

Based on our research questions, we expect to see the following hypotheses come to fruition:

1. The nearest defender in the frame prior to a *slowdown* is most responsible for the ballcarrier's deceleration.
2. Accumulated *slowdowns* will positively correlate with tackles, assists, and defensive value.

[3] Box score data is where counting statistics and some advanced statistics from the game can be seen.

3. Closer defender proximity will significantly predict reductions in ballcarrier speed.

Key Variables

We will focus mainly on the following variables, some of which came directly from the provided dataset. Others we derived, as detailed in the next section.

- Ballcarrier Speed.
- Defender Distance.
- Defender Density.
- Offensive Play Type.
- Frame.

Methodology

In the following subsections we detail our methodological approach in preparing the data for analysis. We start with our approach to data profiling, new variable construction, and descriptions. Lastly, we explain how ballcarrier speed and its changes were measured.

Data Acquisition and Profiling

We utilized the pandas [7] Python [8] package in Jupyter notebooks [9] to read all the Comma Separated Value (csv) files. They were titled **games, players, plays, tackles**, and they included 9 weeks of **tracking** data (each week with its own file). We began by using the y-data profiling [10] package for a data profile overview for each dataset. This gave us insight into the shape of each attribute and variable in addition to summary statistics. We share specific findings from the data profiling in section 4. A check was done to remove all nulls, or blank records, via custom helper functions.

Similar custom functions were used to remove duplicate records. We checked for consistency within a single file as well as across files. A composite primary key was made from four of the key identifier variables (*gameid*, *playid*, *nflid*, and *frameid*) for the database. These combined variables were chosen because they satisfy the condition of being boiled down to a unique observation. While there will be some rows with the same *gameid*, using all four variables forces a lack of duplicates. This combination of four makes for a very granular dataset, where we know what game, what play, which player, and each frame in the play.

There were many missing "NFL IDs" in the tracking data. Those were for the football itself, so a zero was assigned as its ID. There were seven players that had an inconsistent birthdate format. That was corrected to align them with the rest. After auditing the. csv files we made a new dataframe with all of them combined and a new consolidated data profile report was ready for further analysis.

Creation of the Data Model

The next step was to create the data model. We transformed all data to their proper type in terms of strings, numeric, dates, times, etc. We defined database tables with the Structured Query Language (SQL) Data Definition Language (DDL) [11]. Finally, we loaded the tables into the data model and ended up with over twelve million observations ready for analysis.

Describing the Data

The twelve million observations were made up of 13 smaller datasets, amounting to 17 variables across 136 games. The weeks contained 15.1 games, on average. The reason that it was not 16 per week (as expected since there are 32 NFL teams) was because there are scheduled weeks off for each team, known as a BYE week.

The number of plays that we had in the dataset was 12,483. Over the course of 136 games, this amounted to an average of 91.78 per game, where each team would get about 45.89 plays per game, on average. We found 480 unique ballcarriers within the dataset, good for an average of 15 per team over the first nine weeks of the season. This was surprising at first. However, after considering injuries, special personnel packages, and rotating players, it began to make more sense as NFL teams are known for spreading the ball around more often in the modern game than in years past.

The down and distance data were also included in the **plays** dataset, along with data on pass lengths and the results. The offensive formation data paired with how many defenders were in the box on any given play were also interesting variables to consider. Lastly, each play's expected points added (EPA) was also included [12]. EPA is a relatively new, advanced metric that gives an idea of what the expected outcome might be for any given type of play, in terms of how many points we can expect to be added by that play type.

One of the smaller datasets that made up the whole was the **players** dataset. This dataset contained *height*, *weight*, *birthdate*, *college*, *position*, and *name* of each player in the games. There were 1,683 players in all. This amounted to 52.59 players per team, which is roughly in alignment with expectations since the NFL has a 53-man roster limit.

The next ten datasets were vital to our analysis. The first was the **tackles** dataset. This one contained 17,426 observations where each game, play and player were identified and whether or not they made a full tackle or assisted on a tackle.

Thanks to PFF, missed tackles were also recorded. The idea for our new advanced statistic, *slowdowns*, originated from the fact that tackle data is so limited. Counting statistics are binary, and therefore, basic. Our idea was to contribute to this area of football as we believe that it was previously underdeveloped. This is often the case in sports, as defensive statistics are typically harder to record and measure than offensive ones. This is due to the fact that offensive statistics are typically tied to measurable outcomes like points scored, catches made, or yards gained.

Lastly, there were nine datasets of **tracking** data, with each of them containing a week's worth of games. In total, they comprised 1.4 million observations. The reason that the dataset was so large is because it was broken down by *game*, *play*, *player*, and *frame*. With 136 games, 91 plays per game, 22 players on the field being tracked at any given time along with the officials and the football, and up to 60 frames per play, the dataset was quite granular and added up quickly. Each frame captured over 10 records per second, recording each player's location, speed, distance traveled, and more. You can think of a frame like a frame in a film—a discrete moment in time, when the sensor data was captured. This is somewhat analogous to the concept of digital snapshots versus analog motion.

Distance Matrix

Each individual players exact X and Y coordinates on the field could be measured thanks to the Next Gen Stats tracking data. The tracking data also recorded the *direction* the play was headed (left or right), the *orientation* of each player on the field (0-360 degrees), the *speed* and *acceleration* of each player, and the *event* (handoffs, passes, results, etc.).

To move the data forward and build a key building block for our analysis, we created a distance matrix, based on each player's X and Y coordinates on the field in each frame. The distance matrix measured each defender's distance from the ballcarrier during each frame. We used the Euclidean Distance Formula to find this. By summing the squared differences between the X coordinates for the ballcarrier and defender A, as well as the Y coordinates for those two, respectively, we then just had to take the square root to find the Euclidean distance from the ballcarrier and any defender at any given time in the game. Figure 1 below outlines the relationship between ballcarriers and defenders:

The Euclidean Distance Formula is closely related to the Pythagorean Theorem. If you consider that the difference in X coordinates mentioned above could simply be referred to as *a*, and the difference in the Y coordinates could be looked at as *b*, it becomes clear that we are triangulating and simply doing the Pythagorean Theorem:

$$a^2 + b^2 = c^2$$

To find the straightforward distance between the ballcarrier and defender of interest, we just had to take the square root of both sides, arriving at *c*. For the sake of the Euclidean Distance Formula below, we use *d* for the defender distance.

$$d = \sqrt{(x2 - x1)^2 + (y2 - y1)^2}$$

The next subsection details how we progressed from this point by focusing on the combination of defender distance, and speed.

Speed Curves

The second main ingredient to our approach was ballcarrier speed data. The idea was to track the ballcarrier's speed in terms of yards per second (yds/sec), as well as miles per hour[4] (mph) from the beginning of the play until the end. We focused on the apex of speed on any given play and aimed to track the path down until the end of the play.

Input Data

Player	x	y
A	1	2
B	2	5
C	3	4
D	5	5
E	5	4

Output Data: Distance Matrix

	From Player				
To Player	A	B	C	D	E
A	0	3.162278	2.828427	5	4.472136
B	3.162278	0	1.414214	3	3.162278
C	2.828427	1.414214	0	2.236068	2
D	5	3	2.236068	0	1
E	4.472136	3.162278	2	1	0

Fig. 1 Our distance matrix, using example values to illustrate how it functions

[4] Since the USA is not on the metric system, miles (1.609 km) per hour is a common form of measuring speed.

Fig. 2 This image is a representation of a single frame from an entire play where Buffalo Bills Quarterback[15] (QB) Josh Allen has passed the ball to his Running Back[16] (RB), Devin Singletary, who evaded the defense (represented in yellow) for some positive yardage before being caught by the defense (represented in orange). Each player is represented by their jersey number in the circles, and the black arrows denote the direction that they are facing, thanks to the tracking data. This was Game #2,022,090,800 and Play #122

[15] The QB in gridiron football is a position that is viewed as the most important, as they are the conductor of the offense. As Hall of Famer Howie Long wrote "Being a former defensive player, I hate to admit that the quarterback is the most important player on a football team." [14] They are responsible for the distribution of the football to different players and leading the team down the field into the end zone.

[16] The RB is mostly tasked with carrying the football and running forward for as many yards as possible on any given play. They also have responsibilities of blocking at times, as well as receiving the football via a pass through the air.

For an example of ballcarrier speed on one play, consider this completion[5] to Devin Singletary of the Buffalo Bills in the 2022 season opener against the Los Angeles Rams. Figure 2 shows a screenshot from the visualization we made [13] of the entire play:

[5] A completion occurs when a forward pass is transferred from one player to another and caught successfully.

Figure 3 shows Singletary's speed by frame during the play. His speed gradually accelerates at first, and tops out at 12.8 mph. From there it is all downhill.

Our approach was to measure the decreases in speed across the frames after peak speed. Figure 4 shows that looked like on this particular play:

Each of these decreases in speed, by frame can be isolated and quantified. Since we wanted to assign credit for them, we brought in the defender distance data as well. Defensive players should be given credit for causing a decrease in ballcarrier speed, however small the decrease might be. Just like a tackle or an assisted tackle can be recorded, we measure slowdowns for each defender's statistics on any given play, game, or across many games.

We found the distances of all eleven defenders for every frame for all plays in weeks 1–9 of the 2022 NFL season. By focusing on the post-peak ballcarrier speed, we measured the slowdowns and who should be given credit for each. We assumed that the nearest defender who caused the ensuing slowdown should be given credit for that slowdown. The logic being that there is a natural lag effect. If a slowdown occurred in one frame the assumption is that the nearest defender in the previous frame was the one who caused the ballcarrier's speed to decrease. This assumption will be tested in section 5.

Assigning Slowdown Credits Using Example Plays

In the Singletary play, speed decreases in mph and yds/sec were tracked, and so were slowdowns. Thus, each defender who was the closest in terms of distance was given credit for their ballcarrier slowdown contribution for the next frame, so long as it was after peak ballcarrier speed.

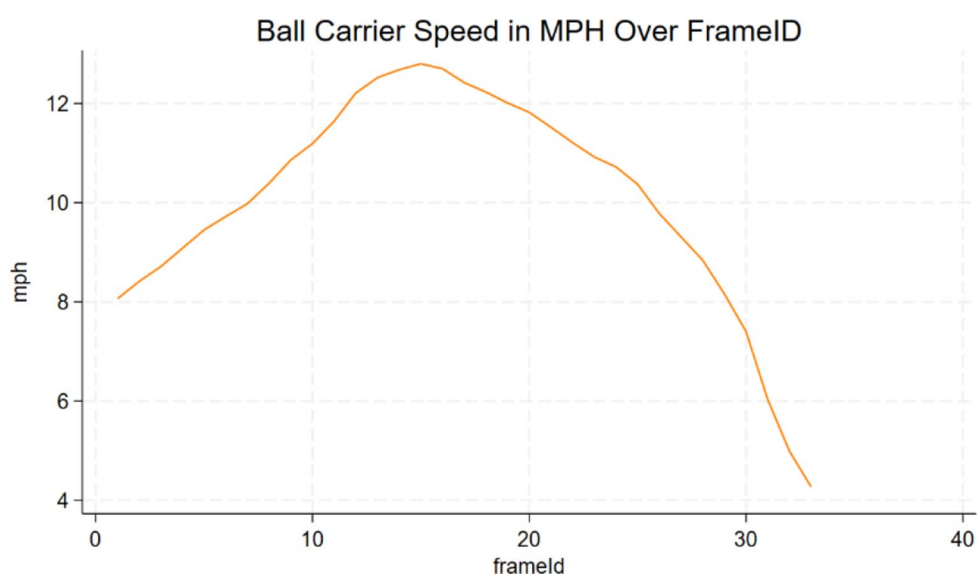

Fig. 3 This image represents what a fairly typical speed distribution would look like for a ballcarrier on a typical offensive play. In this case we are measuring Singletary's speed on play #122 of Game #2,022,090,800

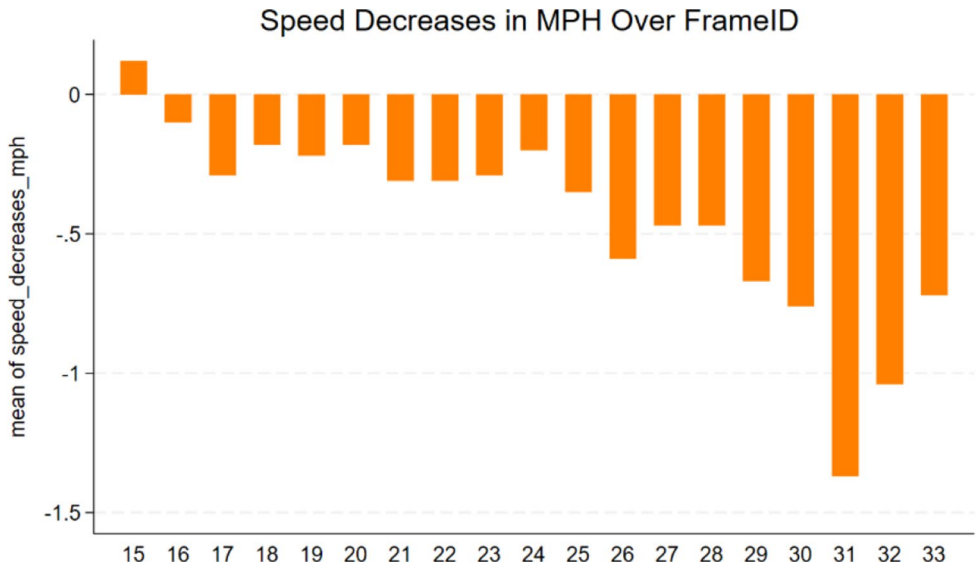

Fig. 4 Although the overall trend in this visual shows that decreases in speed were growing larger over the course of the play, the sharpest decrease was near the end, which would represent a typical play as speed should drop sharply to near zero just before a player is tackled

The defender who made the largest contribution on that play was Linebacker[6] (LB) Bobby Wagner. He contributed a speed decrease of 1.37mph on that play. In this case, Wagner did make the tackle, but that is not always the case.

In the next example play from the Week 1 game between the San Francisco 49ers and Chicago Bears, the description of the play from the play-by-play data on Pro Football Reference [15] is "Justin Fields pass complete short middle to David Montgomery for 16 yards (tackle by Tashaun Gipson)". However, when watching the game tape from that play you can see multiple San Francisco 49er defenders impacted the speed of Bears RB Montgomery, not just the 49ers Safety[7] Gipson.

This is a perfect example of where a slowdown credit can highlight a contribution that would be missed when solely reviewing box score data. Fred Warner had the biggest slowdown at frame 57 and decreased the ballcarrier's speed by 1.62 miles per hour.

Defensive Tackle[8] (DT) Javon Kinlaw as well as LBs Dre Greenlaw and Fred Warner also had a hand in bringing down Montgomery by getting near him. None of them were credited with an assist. Instead, Kinlaw and Greenlaw recorded missed tackles. By leveraging our slowdown metric, coaches could give them credit in addition to the tackle for Gipson.

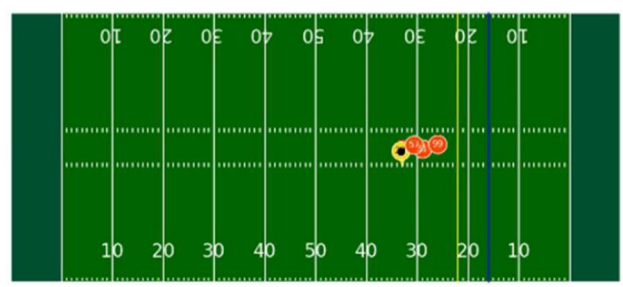

Fig. 5 This figure highlights only key defenders in Play #364 in Game #22,091,102. In short, the three defenders chosen were the ones who contributed the most to making the stop

Figure 5, a screenshot from our second animation [16], highlights the players who were close to Montgomery all throughout the play. By parsing out the other eight defenders it becomes easier to see the three main defenders who contributed to bringing Montgomery down when viewing the full animation:

Descriptive Analysis

In this section we will shift our focus to analyzing trends in an aggregate sense, looking for meaningful patterns and predictive power.

Correlations

We have gone over a couple of plays now as examples, which can be helpful for coaches' evaluation of players in a micro sense. Now, we let patterns and trends emerge by reviewing slowdowns in a macro sense over 9 weeks of

[6] The LB is a defensive player usually tasked with making the most tackles on the defense as they are in the middle of the field. They are meant to stop running plays, and at times assist in pass defense as well, making them versatile members of the defense.

[7] A Safety is usually in the very rear of the defense, as they are the last line of defense preventing opposing ballcarriers from entering the end zone.

[8] DTs are typically the largest player on the defense, with their main responsibility being to create pressure at the point of attack for an offense. They are as close as possible to the ball at the beginning of each play and they use their size and strength to occupy blockers and stymying the offensive gameplan from the beginning of each play.

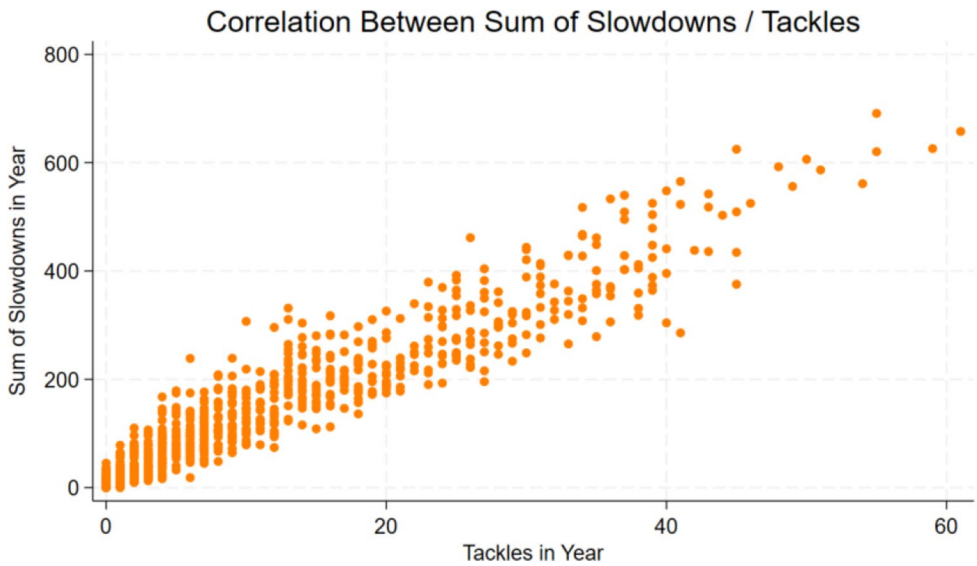

Fig. 6 This scatterplot shows the correlation between sum of slowdowns and sum of tackles over the first nine weeks of the 2022 NFL season. A strong, positive correlation has been revealed

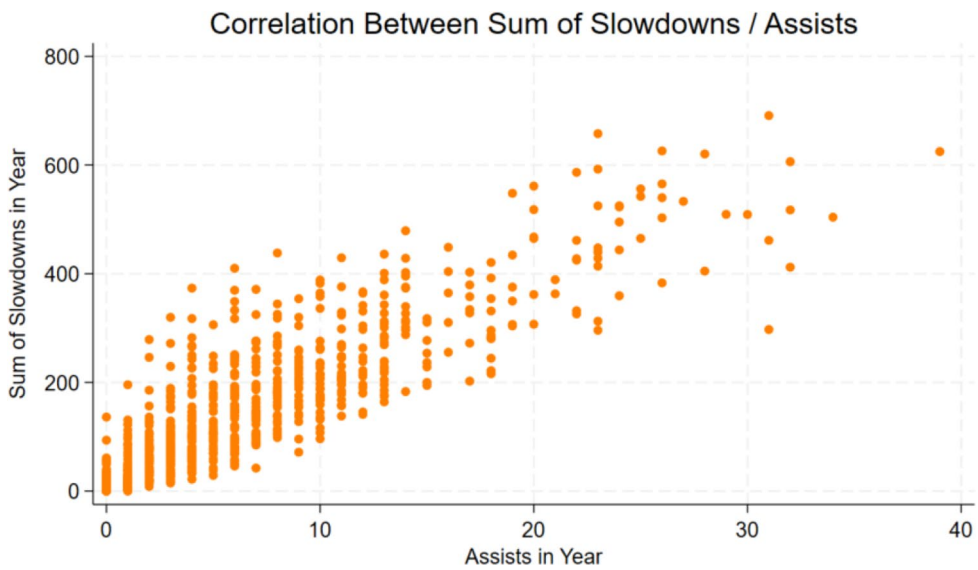

Fig. 7 This scatterplot shows the correlation between sum of slowdowns and sum of assisted tackles over the course of the first nine weeks of the 2022 NFL season. A strong, positive correlation is shown, but it is weaker than the relationship between the sum of slowdowns and sum of individual tackles across that timeframe

games. By this point outliers would not have as strong of an effect on the results, leading to more accurate assessments of a defender's abilities.

Here we observe the relationship between the sum of slowdowns and the sum of individual tackles. Figure 6 shows the correlation between total accumulated slowdowns in mph and tackles was 0.92.

Next, Fig. 7 displays the relationship between the sum of slowdowns and assisted tackles. The correlation that we observed was 0.82.

Since we have observed such strong correlations between both tackles and assists by the sum of slowdowns in mph, slowdowns prove to be valuable in assessing a defender's contributions to the team's goals of tackling ballcarriers as soon as possible. Even if a defender does not record a lot of tackles or assists in any one game in particular, it is possible that they had a great effect on the game if they accumulated a lot of slowdowns.

Top Performers

We now show which defenders in the NFL accumulated the most slowdowns across the first nine games of the 2022 season. This demonstrates the value of the slowdowns is not only correlated with both individual and assisted tackles, but also overall high-quality individual defender play. Figure 8 shows the top 10 defenders in terms of slowdowns:

Each of these players is clearly contributing to the defenses in a large way, and they all deserve credit. Although it is unlikely that coaching staffs have not noticed their tenacity in pursuit of ballcarriers, it may be possible that other players on the complete list are hidden gems.

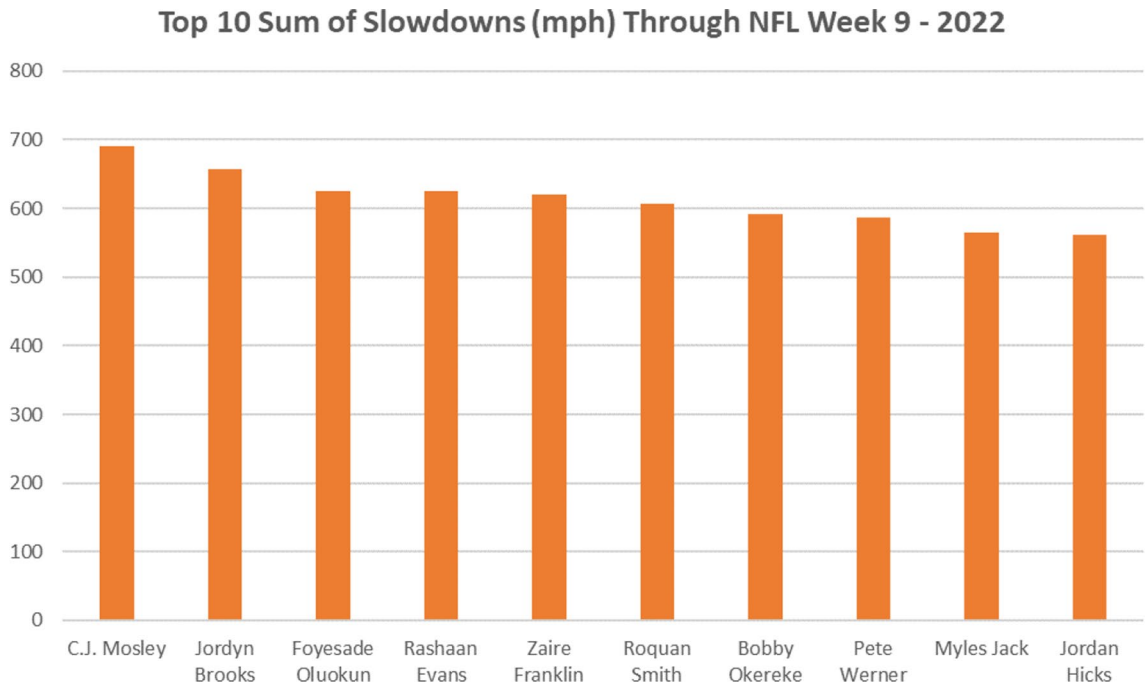

Fig. 8 This bar chart displays the top ten players by sum of slowdowns. C.J. Mosley earned first place with nearly 700 points worth of accumulated slowdowns (in mph) in the span of the first nine weeks of the 2022 NFL season

Table 1 This table outlines the top-ten players in terms of the sum of slowdowns (mph)

Sum of Slowdowns 2022 Weeks 1–9 Rank (mph)	Player	Team	Player Position	Sum of Slowdowns 2022 Weeks 1–9 (mph)
1	C.J. Mosley	New York Jets	LB	691.02
2	Jordyn Brooks	Seattle Seahawks	LB	657.76
3	Foyesade Oluokun	Jacksonville Jaguars	LB	625.95
4	Rashaan Evans	Atlanta Falcons	LB	624.83
5	Zaire Franklin	Indianapolis Colts	LB	620.37
6	Roquan Smith	Baltimore Ravens	LB	606.22
7	Bobby Okereke	Indianapolis Colts	LB	592.47
8	Pete Werner	New Orleans Saints	LB	586.51
9	Myles Jack	Pittsburgh Steelers	LB	565.25
10	Jordan Hicks	Minnesota Vikings	LB	561.30

Table 2 This table retains the sorting from Table 1, but now displays tackling data

Total Tackles 2022 Rank	Player	Total Tackles	Games Played	Tackles per Game
9	C.J. Mosley	158	17	9.29
6	Jordyn Brooks	161	16	10.06
1	Foyesade Oluokun	184	17	10.82
7	Rashaan Evans	159	17	9.35
4	Zaire Franklin	167	17	9.82
3	Roquan Smith	169	17	9.94
10	Bobby Okereke	151	17	8.88
99	Pete Werner	80	12	6.67
48	Myles Jack	104	15	6.93
18	Jordan Hicks	129	17	7.59

To say that the players who made the top ten sum of slowdowns list are quite accomplished would be an understatement. C.J. Mosley and Roquan Smith made the 2022 All-Pro team[9]. Tables 1 and 2 show that nine[10] of the top ten sum of slowdowns leaders also ended up making the top ten in total tackles by the end of the season. It is safe to say that

[9] An All-Pro player [17] is one who was selected by the media as the best at their respective position for that year. It is a very prestigious award due to the scarcity and the fact that fan voting is not considered.

[10] Except for Jordan Hicks of the Minnesota Vikings, who finished 18th with 129 tackles.

as long as a player stays healthy, slowdown data relates to tackles and therefore, contributions to the defense.

Not surprisingly, we found that all members of the top-ten in terms of sum of slowdowns were LBs. This position group is generally placed in the center of the defense and is responsible for being involved in almost every play, whether it be against the run or pass. Due to this responsibility, it is logical that LBs would be provided with ample opportunities to rack up lots of slowdowns over the course of nine weeks.

Although most of the members of the top-ten for sum of slowdowns also finished the season in the top-ten for total tackles, C.J. Mosley only finished in 9th place for total tackles as opposed to 1st for sum of slowdowns. This was despite the fact that he played in all 17 games. Since he earned All-Pro honors for the 2022 season along with Roquan Smith, this suggests that there is inherent value in a metric such as slowdowns, as it captures what the eye is seeing in terms of a player being involved and influencing plays.

In short, individual tackles are shown to not be the end-all be-all. Slowdowns provide a way to quantify the player tracking and defensive influence data that our eyes seek out as fans watching the games or when coaches study film.

Unfortunately for LBs, their position group has been valued less in terms of their salary share relative to other position groups in recent years. This is despite leading the tackle and *slowdown* rankings. Curiously, the Linebacker salary rankings [18] shown here for 2022 did not match well with slowdown data. Experience would not be a factor for the low salaries shown here, as the years in the league listed here would suggest that many of these players are in prime earning years, versus younger players on cheaper rookie contracts.

The likely answer is that Edge Rushers, Defensive Ends, and Outside Linebackers (OLB) have been categorically lumped together[11] as a recent trend in the NFL. Edge Rushers, Defensive Ends (DE), and OLBs in certain types of defensive schemes tend to get paid much more than Interior Linebackers (ILB). Given most players on this list are not Edge Rushers or OLBs, that could explain the discrepancy in pay.

Other factors for pay would include draft position, scarcity, or team needs at the time. Player contracts are structured so that the position in which the player was drafted matters in terms of what they can expect to be paid, moving from different scales depending on when they were selected by their teams in the NFL Draft[12] or in the Undrafted Free Agent (UDFA[13]) pool. Teams hope to discover "diamonds in the rough" when it comes to finding talent in the UDFA pool that all teams passed on, originally.

A diamond that slowdown data reveals is Pete Werner of New Orleans. After the first nine weeks he finished 8th in sum of slowdowns, but he only had two years of experience, so in terms of LB pay he was ranked #141 for the year. A profile like this could indicate that this sort of player to watch out for future success at the LB position. Future salaries may hinge on slowdown ability assuming that they can stay healthy.

Given the 0.82–0.92 correlation between slowdowns and assisted or individual tackles, combined tackles can be a proxy for slowdowns. Werner's #74 overall ranking in combined tackles across *all* position groups was quite impressive for the 2023 season [19]. Entering the 2024 season [20], Werner will earn $1.43 M, an increase of about 57%, landing him the #69 ranking for LB base salary pay. Werner provides tremendous value relative to his production.

Slowdowns by Position

Summary statistics for the other position groups in terms of slowdowns through the first nine games of the 2022 NFL season are shown in Fig. 9:

The table is grouped by position groups in a very granular sense, with a distinction between Cornerbacks (CB) and Safeties, and with even more granularity between Strong Safeties (SS) and Free Safeties (FS), for example.

The numbers made intuitive sense, as we expect a SS to be more involved in run defense than a CB, hence the higher mean for the slowdowns in comparison. The granularity of position groups allows us to observe differences in SS versus FS, as well as DE versus DT, instead of just lumping them all in together as Defensive Backs or Defensive Lineman.

This way we are not unfairly assessing a DE to a DT without considering context. What we hope that coaches do is compare apples to apples, which in this case would be DE to DE, DT to DT, and so on. If the *ceteris paribus* assumption holds, valuable insights about a player's hustle mentality and effort can be gained at the margins of a statistical analysis of their slowdowns.

[11] Edge Rushers are players who are primarily tasked with lining up on the outer edge of a defensive front, facing the offensive line and near the line of scrimmage. They are tasked with applying pressure on opposing QBs, and setting the "edge" so that they can contain ballcarriers within their own or their teammates reach.

[12] The NFL Draft is where incoming players are selected by the 32 active teams. There are seven rounds, with the best players being selected in the earlier rounds. Teams can trade their picks for players or other picks in the same draft or future drafts.

[13] Undrafted Free Agents are incoming players who were not selected in the NFL Draft. Instead, they are a part of the open market player pool. They are free to sign with any team that they wish, instead of being selected by a team, albeit to a lower wage compared to their drafted counterparts.

Sum of Slowdowns (in mph) Through First Nine Weeks, 2022

Position	Count	Mean	Std Dev.	CV	Max
CB	191	142.32	113	0.79	509.24
FS	78	170.31	140.65	0.83	509.03
SS	64	203.24	146.99	0.72	502.89
MLB	11	177.84	162.07	0.91	427.62
ILB	89	255.31	213.2	0.84	691.02
OLB	140	128.54	135.35	1.05	620.37
DE	125	89.96	71.8	0.80	326.9
DT	127	108.57	82.71	0.76	379.39
NT	31	132.87	83.09	0.63	331.37

Fig. 9 This visual leverages heat mapping to identify higher means and maxes in green, and lower standard deviations and coefficients of variation in green, where green represents "good" and red is "bad". The position groups are as follows: Cornerback (CB), Free Safety (FS), Strong Safety (SS), Middle Linebacker (MLB), Inside Linebacker (ILB), Outside Linebacker (OLB), Defensive End (DE), Defensive Tackle (DT), and Nose Tackle (NT)

It also is intuitive that more central positions to the defense tend to do better in terms of slowdowns, as Safeties posted better means than CBs, DTs did better in terms of mean slowdowns than DEs, and so did the ILBs compared to OLBs. Those centered positions have more access to either side of the field in terms of their sideline-to-sideline defense of the field. If an offensive play is run to the opposite side of the field of a certain CB or DE, the likelihood that they are the one to have the slowdown awarded to them is low.

This concept truly highlights why slowdowns tend to stand out for LBs, since they are in the middle of the action and the center of the defense, especially ILBs. However, this does not stop slowdowns for being a useful metric for any position group, as long as the comparisons between players are fair. We believe that the correct starting point would be for them to belong to the same position group, with more granularity or specificity leading to the best results when comparing and contrasting.

Empirical Validations and Further Analysis

In this section we put our hypotheses and assumptions to the test by attempting to validate them. Our goals included fully explaining our logic and reasoning in a step-by-step manner, quantifying relationships, and looking to see if what we believed to be true correlated with the reality of the data.

Lag Assumption Testing

Figure 10 shows how the assumption of a one-frame lag held up versus two and three-frame lags. The distance of the closest defender from one frame ago better explained current ballcarrier speed relative to two and three frames ago, in terms of R^2.

We also ran a regression as an added layer to further illustrate the effect of lags and how a one-frame lag was the best choice for the analysis. This time, the absolute value of a slowdown in miles per hour (*abs_val_slowdown_mph*) was the dependent variable. The independent variables were the distance of the nearest defender either in the previous frame, previous second frame, or previous third frame. In Fig. 11 you will find the resulting coefficients from the three separate single predictor variable regressions, where they all were statistically significant.

The key point of this graph is to illustrate that a one-frame lag was once again, the best choice to help explain ballcarrier speed and changes to it in the current frame. In this case, the inverse relationship was expected. As the distance that the nearest defender was to the ballcarrier decreased, the expected slowdown of the ballcarrier's speed increased. This helps to demonstrate the importance of defender proximity, especially in the previous most recent frame compared to older ones.

Further Analysis

We took our statistical analysis further by using more inferential statistics, specifically frequentist probability with

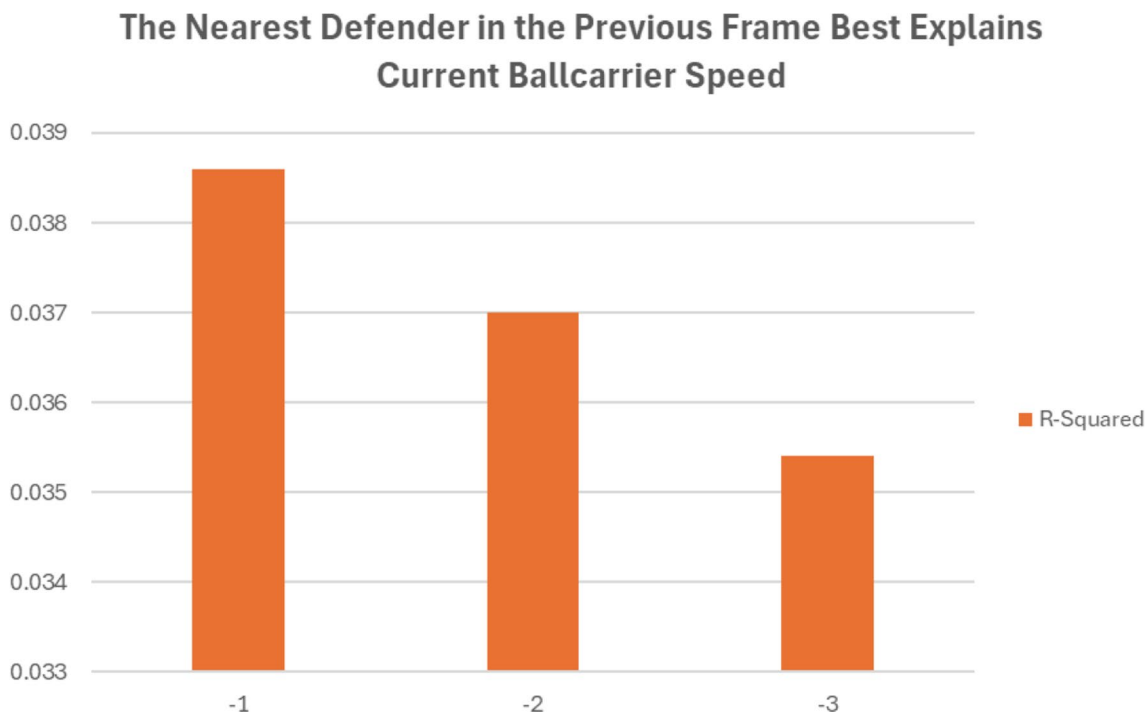

Fig. 10 This bar chart has the R^2 on the y-axis and a one-frame, two-frame and three-frame lags on the x-axis

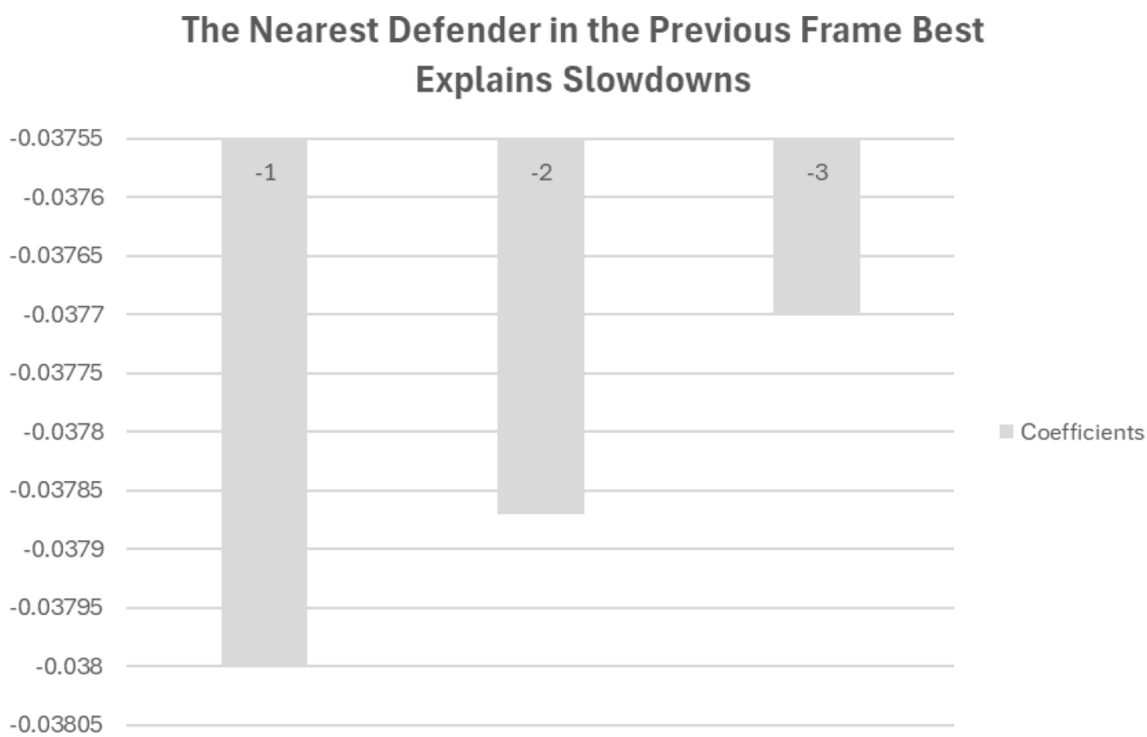

Fig. 11 This bar chart has the coefficients on the y-axis and a one-frame, two-frame and three-frame lags on the x-axis. Although the resulting coefficients from the three single univariate regression models were small, they were all statistically significant

different variables. We sought validation of our hypothesis that having defenders closer to the ballcarrier impacted the ballcarrier's speed. Specifically, we looked to see if there was a statistical relationship between the distance a defender is from the ballcarrier and the ballcarrier speed in yards per second. We calculated the Pearson correlation coefficient between speed and defender distance within one yard.

In the Fig. 12 scatterplot there are 283,336 observations plotted and we found a very weak (although significant) positive correlation between speed and defenders less than 1 yard away r(283,334) = 0.092, p <.001. Thus, as the defender distance increases, so does ballcarrier speed. This aligns with our hypothesis, with less defensive players nearby, the runner is free to accelerate. Conversely, as defender distance decreases, this correlation implies the speed of the runner would follow and decrease. In this case we aimed to isolate the relationship between one ballcarrier and one defender. Given the fact that there are eleven defensive players on the field for each play, each with the ability to influence the ballcarrier's speed and path in some way it becomes understandable that a one-on-one relationship is challenging to capture.

However, the aim of this paper was to assign credit to individual defenders so that they could have a new variable, *slowdowns*, shown alongside the usual tackling statistics for further context and nuance as to their ability to contribute to their respective defenses.

Table 4 lists the results from an Ordinary Least Squares (OLS) regression with distance under one yard as a predictive variable, which was found to be statistically significant. The coefficient was 0.75 and the p-value was less than 0.00. So, we found that as the defender's distance away from the ballcarrier decreases by one yard, the ballcarrier's speed is expected to decrease by 0.75 yds/sec. This can be the difference between a ballcarrier earning a first-down or touchdown, or the defender making a clutch stop. Many coaches like to remind their players that "football is a game of inches". So, every yard, or 0.75 yards in this case, really counts.

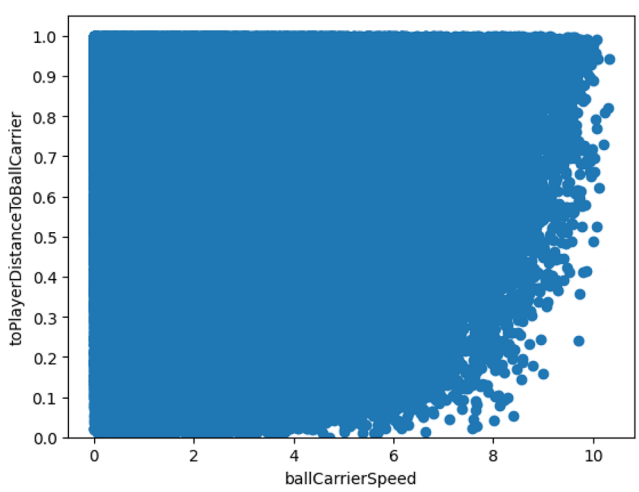

Fig. 12 Correlation between ball carrier speed on the x-axis and distance from the ball carrier on the y-axis

Doubling the distance to defenders less than two yards away again yielded a weak, positive correlation (0.17) and a statistically significant, positive coefficient (0.71). We repeated this for less than three and four yards. Each also had a significant coefficient (3 yards = 0.61 and 4 yards = 0.51, respectively).

The main takeaway here for coaches is that players who are less than four yards away have a significant impact on slowing down the ballcarrier. A typical saying that coaches say is to "swarm" the ballcarrier and this is why that matters. Even if the defenders are on the backside of the play, they can reduce any other options for the ballcarrier by increasing proximity as soon as possible. This is why everyone praises defenses where they see "11 shirts" headed towards the ballcarrier right away on a consistent basis.

Analysis by Type of Play

In addition to seeing how the statistical relationship between runners and defensive players changes as distance increases, we also tested how it fluctuates based on the type of play. For the analysis above we grouped all of the plays together, but as you can see in Fig. 13 below, over 66% were designed rushes:

Looking only at those designed rushes where the runner (often a RB) was handed the ball (typically by the QB), we found that the defenders less than two yards away predictor was significant but only explained 3.9% of the variance in the model (R^2 = 0.039, F(1, 435274) = 17,740, p <.01).

The regression equation was ballcarrier speed = 1.7846 + 0.7859 * defender distance and the distance coefficient was significant once again (β = 0.7859, p <.001). The coefficient was slightly higher than for all play types in Table 4. We had the same finding with pass completions. In the same subset of data with defenders less than two yards away, the coefficient was smaller (β = 0.6448, p <.001) implying that individual defender distance is less impactful on Wide Receiver[14] (WR) speed after they catch the pass, relative to Running Backs. In this case for defenses, it is even more important to swarm the ballcarrier as each single defensive player has less impact. For this type of play the single independent variable predicted just 2.2% of the variance in the model (R^2 = 0.022, F(1, 158402) = 3,600, p <.01).

Scramble plays accounted for only 3.34% of all plays. For most modern offenses, this is indeed the least common play

[14] WRs typically line up near away from the QB before the snap closer to the sidelines and opposing CBs. Their main responsibility is catching passes from the QB.

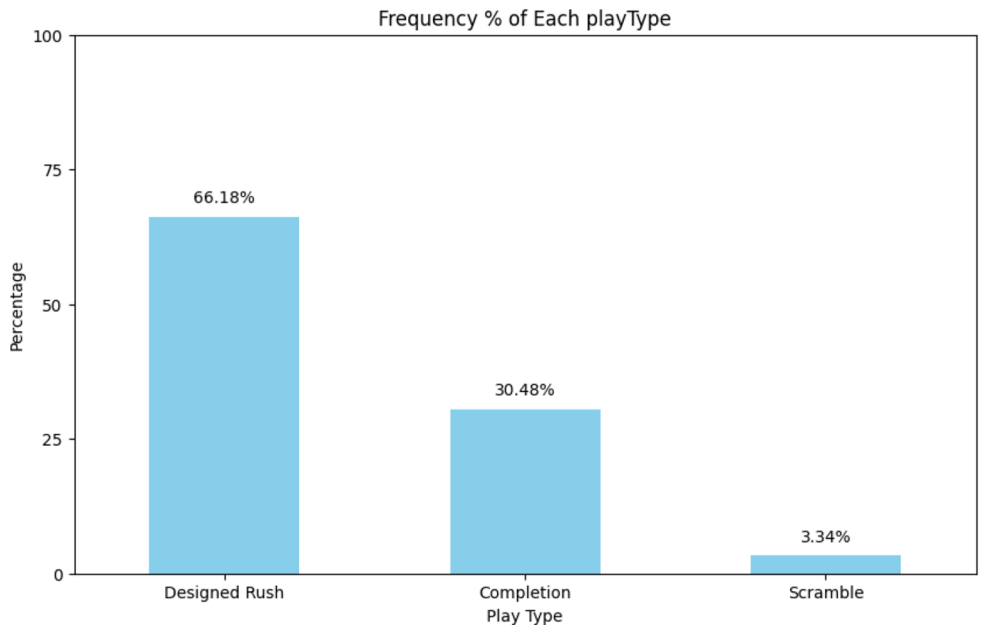

Fig. 13 The distribution of the 12,483 plays. The percentage is on the y-axis and three types of plays are on the x-axis

type. Some QBs are not very effective at running and prefer to stay in the "pocket" protected by the offensive line. An example of a quarterback who is very comfortable scrambling is Lamar Jackson. His prowess at this type of play has helped him win two Most Valuable Player (MVP) awards. He is the exception and most QBs are less talented at carrying the ball than RBs and WRs. That is likely why the coefficient for defenders less than two yards away on scrambles is the biggest among all three play types ($\beta = 0.9447$, $p < .001$). It is the highest coefficient we have seen among all the univariate regressions predicting ballcarrier speed. The variable explains 4.4% of the variance in the model ($R^2 = 0.044$, $F(1, 8834) = 405$, $p < .01$). Across all play types, we are most confident in our finding that defender distance influences the ballcarrier speed on scrambles. In the next section we expanded our model to include additional variables which yielded improved predictive results.

Play Type as Part of a Multiple Regression Model

We went deeper into analyzing by play type by using the options directly as independent variables in a multivariate OLS regression model. Dummy variables were made for designed rushes and scrambles. The interpretation of those coefficients compares them with completed catches as the reference group. The model also uses defender distance as an independent variable and, same as the other models, ballcarrier speed was the dependent variable.

The remaining variables were: the number of defenders within 5 yards, Frame ID, a dummy variable for if the play included a missed tackle, an intercept and controls for every Game ID (134 total). We restricted the dataset to only include plays where the closest defender was within 2.5 yards to understand specifically how ballcarrier speed is impacted with nearby defenders. As we saw in Table 4, there is a drop in coefficients for players that are 3 or more yards away compared to 2 yards or less. There were 754,558 records remaining. This led to a substantial improvement as 40.1% of the variation in speed the was explained by the model ($R^2 = 0.401$, $F(141, 754416) = 117.8$, $p < .01$).

We did Variance Inflation Factor testing for this model for the non-fixed effect predictor variables. All had VIF under 5, indicating low multicollinearity and that the model was valid for interpretation:

- Distance to Ballcarrier: 1.011.
- Defenders within 5 yards: 1.279.
- Frame ID: 1.262.
- Missed Tackle: 1.021.

The distance to ballcarrier coefficient was lower here than in previous models (+0.5 yds/sec compared to under 3 yards in Table 4), but the direction was still positive. As an individual defender gets further away the ballcarrier speed increases. Inversely, the ballcarrier speed decreases by -0.7 yds/sec every time there is an additional defender within 5 yards. Ballcarrier speed also decreases as each frame increases, holding all other predictors constant. These findings make intuitive sense, and they quantitatively support our finding that defenders need to work together to slow down ballcarriers, which can be seen in Fig. 14. Speed decreases even more as additional defenders get closer.

When watching games if we see a ballcarrier miss a tackle attempted by a defender, their speed is boosted by the sudden freedom of movement. With the model we validate that observation and show it increases by +0.5 yds/second.

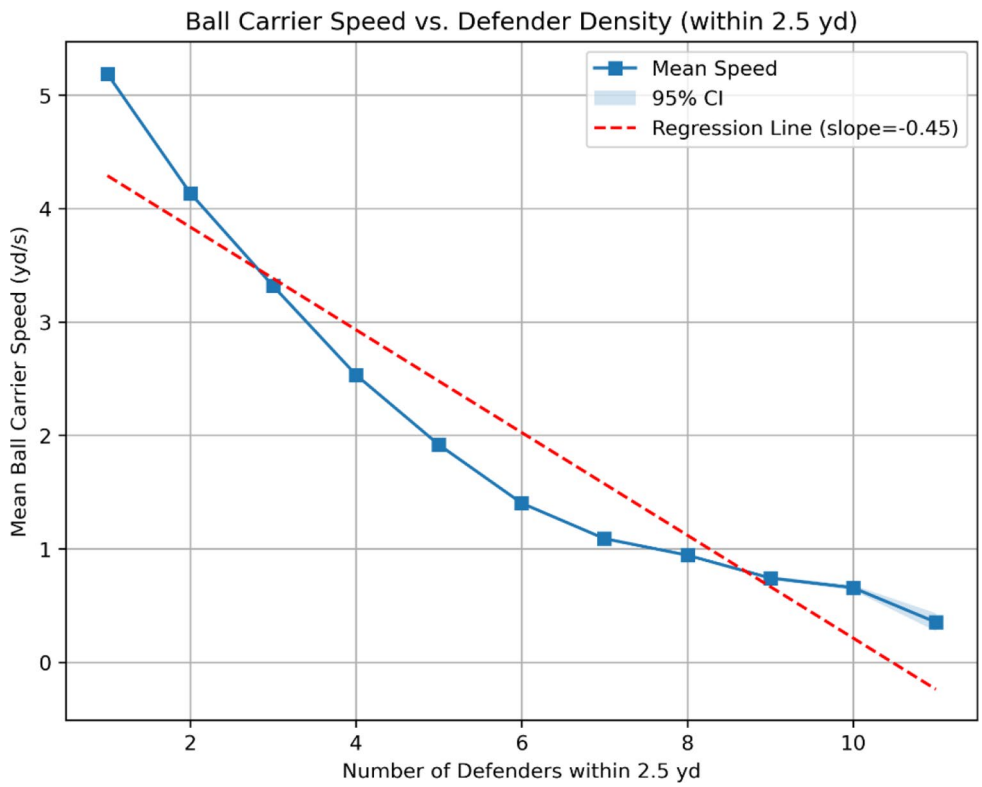

Fig. 14 Chart showing the ball carrier speed on the y-axis and number of defenders on the x-axis. Note the regression line has a different slope than the multiple regression model (-0.660) here because it is only showing the raw, unadjusted relationship between number of defenders and ballcarrier speed

4.2% of this data subset included a missed tackle. Given the end of a football play is either a score by the offense or stop by the defense, Frame ID having a small negative coefficient (-0.007) makes sense. As the ballcarrier approaches zero speed at the conclusion of play, speed overall decreases with frames increasing, as shown in Figs. 3 and 4. We included this and games as variables in the model to account for their influence.

There was a wide range of positive and negative coefficients for games, including some that were significant. It was intuitive that field conditions and the healthy players for a given contest had their own impact on ballcarrier speed.

The dummy variables results are less intuitive. Looking at designed rushes, those plays have +0.57 yds/sec higher speed than completions, all else equal. This was even more pronounced for scramble plays, which have a+1.09 yds/sec higher speed than completions, all else equal. Normally RBs and WRs are faster than QBs. One example is that none of the fastest player 40-yard dash times on record [21] were from QBs, but did include RBs and WRs. However, this coefficient aligns with what we saw in general that quarterbacks have higher speed times in this dataset.

As can be seen in Fig. 15, scrambles score the highest in terms of plays with higher ballcarrier speeds. Given designed rushes and completions tend to occur near defenders, there is less opportunity to reach top speeds. For example, Receivers often need to slow down to concentrate and catch the ball, especially in traffic. Running Backs have their own traffic to work through first in terms of the number of defenders in the box on a designed rush.

Our interpretation of the results is that a scramble occurs after a quarterback scans the field and finds an ideal running lane that has a higher chance of success than a pass. Thus, the defense is more spread out when they do start to run, allowing them to hit higher speeds. The chart only shows plays where defenders are less than 2.5 yards away so that is why speed does not reach the depths that it does in Fig. 3.

Conclusion

Each of our hypotheses from subsection 2.2 came to fruition as outlined below:

1. Figures 10, 11 and 12 demonstrate that the nearest defender in the frame prior to a *slowdown* was most responsible for the ballcarrier's deceleration.

2. Figures 6, 7, 8 and 9, along with Tables 2 and 3 show *slowdowns* positively correlate with tackles, assists, and defensive value.

3. Tables 4 and 5 demonstrate that closer defender proximity significantly predicts reductions in ballcarrier speed.

Based on these findings we recommend that coaches use *slowdowns* in the evaluation of their players. At a strategic level as they make decisions about which 11 players will

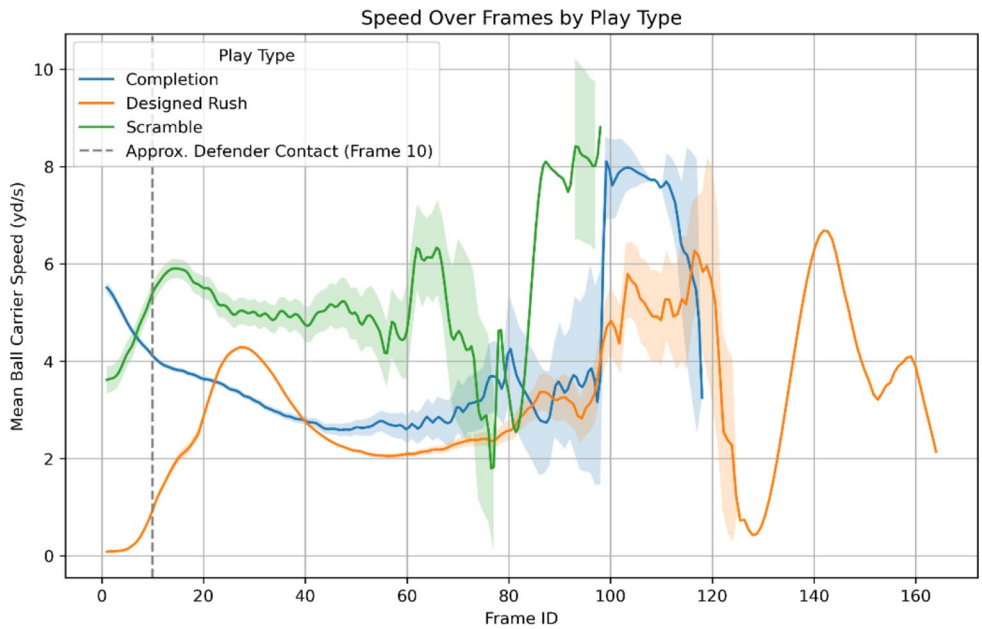

Fig. 15 Line graph showing Frame ID on the x-axis and mean ballcarrier speed on the y-axis

Table 3 This table retains the original order, while displaying salary and experience data

Player	Years of Experience	2022 Base Salary (in Millions)	LB Base Salary Ranking 2022
C.J. Mosley	8	$1.12	69
Jordyn Brooks	3	$1.72	30
Foyesade Oluokun	5	$1.50	35
Rashaan Evans	5	$1.03	87
Zaire Franklin	5	$1.10	79
Roquan Smith	5	$0.57	240
Bobby Okereke	4	$2.54	17
Pete Werner	2	$0.91	141
Myles Jack	7	$1.50	49
Jordan Hicks	8	$1.45	50

Table 4 This table shows the results of statistically testing the relationship between distance from ballcarrier and ballcarrier speed

Distance from Ballcarrier	Coefficient	*P*-Value Significance of Coefficient	Pearson Correlation
<1 Yard	0.7537	<0.001	0.0918
<2 Yards	0.7099	<0.001	0.1711
<3 Yards	0.6063	<0.001	0.2227
<4 Yards	0.5056	<0.001	0.2503

start on their teams, *slowdowns* can help them notice players who are contributing in ways beyond tackles for themselves, but also helping to generate them for others. At a tactical level of evaluating individual plays, we also recommend they highlight *slowdowns* to measure top contributors on both a play-by-play basis when reviewing film, but also in an aggregate sense as the season goes on.

Table 5 This table shows the results of multivariate OLS regression, testing the relationship between distance from ballcarrier and ballcarrier speed

Variable Name	Coefficient	*P*-Value Significance of Coefficient	Standard Error
Intercept	4.3652	<0.001	0.130
Designed Rush	0.5718	<0.001	0.033
Scramble	1.0938	<0.001	0.081
Defenders within 5 yards	-0.6608	<0.001	0.008
Frame ID	-0.0073	<0.001	0.001
Missed Tackle	0.5191	<0.001	0.041
Distance to Ballcarrier	0.4688	<0.001	0.008

During a film session this could be useful to highlight individuals who contributed the most to slowing the ballcarrier down. This would be especially true if that top *slowdown* contributor did not actually end up making the tackle or an assisted tackle. An “unsung hero” would get their due credit.

Additionally, seeing that proximity significantly predicts reduction in ballcarrier speed, we recommend coaches encourage their players to swarm the ball even more fervently. The players may not think it matters if they are too far to attempt a tackle, but even if they are four yards away, it is helpful. Potential cutback lanes or ways for the ballcarrier to have an escape route would be taken away as options. Increasing the number of nearby players helps to get stops sooner.

Player agents are another group that we recommend leveraging *slowdown* data to demonstrate the contributions that the player they represent has made to the team. For the agents representing the next Pete Werner, it could be the difference between a one-year deal, or their client being

offered a multi-year contract extension. *Slowdowns* provide more negotiation points than traditional statistics.

To further extend this analysis we recommend that analytics departments of NFL teams compare *slowdowns* among players in the same position group to see who stands out. Assuming all else about the players being compared to each other is about equal, it could yield a hidden gem as a leading indicator for future tackling success. They can identify those high-value contributors early on and give them more playing time and sign them to contract extensions.

Those data analysts on NFL teams should also consider combining *slowdowns* with other novel metrics such as Fractional Tackles [1], Big Data Bowl winners [22], and Percent Distance Closed [23]. Given they have access to the full breadth of Next Gen Stats, they can look at results over multiple seasons, rather than just the nine game subset we were given access to. We would be interested in seeing how the statistical relationships we studied would have played out over the course of the entire 2022 season and over a greater span of time as well for a larger sample size, and thus, improved accuracy and predictive power.

Advanced defensive statistics in most sports are notoriously hard to measure and rely on, but we believe our straightforward approach to capturing the relationship between ballcarrier speed and defender distance data finally sheds light to those who assist other players in making tackles by hustling on each play. *Slowdowns* provide insight on "hustle metrics" like "who wants it more" and "who gets after it" best. Those fans screaming from the stands can now acknowledge both the tackler and those who helped along the way by slowing the ballcarrier down.

Appendix

You can find some of the Python code we used for data analysis in our GitHub repository https://github.com/danoff/lending-a-hand-in-bringing-ballcarriers-down.

Videos of plays were via code refactored from this 2021 Big Data Bowl submission "NFL Big Data Bowl 2021: Animating Players Movement" https://www.kaggle.com/code/ar2017/nfl-big-data-bowl-2021-animating-players-movement/notebook.

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Declarations

Competing interests The authors have no competing interests to declare that are relevant to the content of this article.

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